

Exercises week 4

Exercises with tentative solutions

Try to do the exercises first without looking at the solutions

Exercise 1: Partially entangled two-qubit state

Consider

$$|\psi(\theta)\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle, \quad \theta \in [0, \pi/2].$$

Tasks:

- 1 Compute $\rho_{AB} = |\psi(\theta)\rangle \langle \psi(\theta)|$.
- 2 Compute $\rho_A =_B (\rho_{AB})$.
- 3 Find eigenvalues of ρ_A and compute

$$S(\rho_A), \quad S_2(\rho_A), \quad S_\alpha(\rho_A).$$

- 4 Identify θ for which entanglement is maximal.

Hint: ρ_A will be diagonal in $\{|0\rangle, |1\rangle\}$.

Solution 1 (full derivation)

Start with

$$\rho_{AB} = |\psi\rangle \langle\psi| = \cos^2 \theta |00\rangle \langle 00| + \cos \theta \sin \theta |00\rangle \langle 11| + \cos \theta \sin \theta |11\rangle \langle 00| + \sin^2 \theta |11\rangle \langle 11|.$$

Use ${}_B(|ab\rangle \langle cd|) = \langle d| b |a\rangle \langle c|$:

$${}_B(|00\rangle \langle 00|) = |0\rangle \langle 0|, \quad (1)$$

$${}_B(|00\rangle \langle 11|) = \langle 1| 0 |0\rangle \langle 1| = 0, \quad (2)$$

$${}_B(|11\rangle \langle 00|) = \langle 0| 1 |1\rangle \langle 0| = 0, \quad (3)$$

$${}_B(|11\rangle \langle 11|) = |1\rangle \langle 1|. \quad (4)$$

Therefore,

$$\rho_A = \cos^2 \theta |0\rangle \langle 0| + \sin^2 \theta |1\rangle \langle 1| = \begin{pmatrix} \cos^2 \theta & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

Eigenvalues: $\lambda_1 = \cos^2 \theta$, $\lambda_2 = \sin^2 \theta$.

Solution 1: Entropies

Von Neumann entropy:

$$S(\rho_A) = -\cos^2 \theta \log_2(\cos^2 \theta) - \sin^2 \theta \log_2(\sin^2 \theta).$$

Second Rényi entropy:

$$S_2(\rho_A) = -\log_2(\rho_A^2) = -\log_2(\cos^4 \theta + \sin^4 \theta).$$

General Rényi:

$$S_\alpha(\rho_A) = \frac{1}{1-\alpha} \log_2((\cos^2 \theta)^\alpha + (\sin^2 \theta)^\alpha).$$

Max entanglement: at $\theta = \pi/4$, $\lambda_1 = \lambda_2 = 1/2$ so $S(\rho_A) = 1$ bit.

Exercise 2: Measurement entropy vs state entropy

Let the single-qubit state be

$$\rho = p |0\rangle \langle 0| + (1 - p) |1\rangle \langle 1|, \quad p \in [0, 1].$$

Tasks:

- 1 Compute $S(\rho)$.
- 2 Compute measurement entropy H_{meas} for measuring in the Z basis.
- 3 Compute H_{meas} for measuring in the X basis.

Key point: $S(\rho)$ is basis-independent, while H_{meas} depends on the measurement.

Solution 2: Z and X measurements

Since ρ is diagonal with eigenvalues $(p, 1 - p)$:

$$S(\rho) = -p \log_2 p - (1 - p) \log_2 (1 - p).$$

Z basis measurement:

$$p(0) = p, \quad p(1) = 1 - p \Rightarrow H_Z = -p \log_2 p - (1 - p) \log_2 (1 - p) = S(\rho).$$

X basis measurement: $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$, with projectors $\Pi_{\pm} = |\pm\rangle \langle \pm|$. Then

$$p(\pm) = (\rho \Pi_{\pm}) = \frac{1}{2},$$

so

$$H_X = 1,$$

independent of p .

Thus generally $H_{\text{meas}} \geq S(\rho)$, with equality in eigenbasis of ρ .

Exercise 3: Mutual information for a classical mixture

Consider a two-qubit mixed state

$$\rho_{AB} = \frac{1}{2} |00\rangle \langle 00| + \frac{1}{2} |11\rangle \langle 11|.$$

Tasks:

- 1 Compute ρ_A, ρ_B .
- 2 Compute $S(\rho_{AB}), S(\rho_A), S(\rho_B)$.
- 3 Compute $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$.
- 4 Compare to a Bell state, where $I(A:B) = 2$.

Solution 3: Classical vs quantum correlations

Compute reduced states:

$$\rho_A = \rho_B = \frac{I_2}{2} \quad \Rightarrow \quad S(\rho_A) = S(\rho_B) = 1.$$

The global state has eigenvalues $\{1/2, 1/2, 0, 0\}$, hence

$$S(\rho_{AB}) = 1.$$

Therefore

$$I(A:B) = 1 + 1 - 1 = 1.$$

Interpretation:

- This state has correlations, but they are purely classical.
- Bell states have maximal mutual information $I = 2$, reflecting strong quantum correlations.